Evaluation modeling of the main flight display interface in the cockpit of the Boeing 747 airplane based on fluency theory

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Abstract

Based on the continuous development of artificial intelligence (AI) in interface recognition technology, the purpose of this article is to build a PFD interface fluency classification model, providing fluency level evaluation for future interface optimization design. The article introduces the influencing factors of fluency theory on the interface of the Boeing 747 PFD (Main Flight Display), and explores the effects of low, medium, and high fluency levels on participants browsing PFD interface information. By optimizing three interface factors, including font size, color contrast, and graphic outline, different levels of smoothness were constructed. Experiments were designed to collect participants' reaction time, accuracy, blink frequency, and subjective evaluations for tasks with different fluency levels. Finally, an interface fluency recognition and classification model was established using the Fisher discriminant method, and its effectiveness was verified. This model replaces the traditional single subjective emotional evaluation and forms an intuitive and quantitative interface design evaluation method. This method provides designers with a new evaluation approach, breaking free from the limitations of subjective evaluation.

CCS Concepts

• **Human-centered computing** → Human computer interaction; HCI theory, concepts and models; Human computer interaction; Interaction paradigms; Graphical user interfaces; • **Theory of computation** → Logic; Verification by model checking.

Keywords

Interface Evaluation, Fluency Theory, Image Classification Model, Aircraft Cockpit Interface

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1 Introduction

The primary flight display (PFD) page serves as the pilot's primary window for accessing and analyzing flight data, including navigational data and important flying parameters. When a pilot is in the cockpit of a civil aircraft, their eyes are frequently fixed on this interface. The PFD serves as a crucial point of reference for pilots, whether the aircraft is landing, cruising, climbing, or descending. As technology advances, the PFD interface's design also changes. It is possible to guarantee that the design satisfies the real operational requirements by taking into account the pilot's requirements for new technologies and interfaces [1]. The PFD interface offers vital information and can be utilized as a decision-support tool to assist pilots in making decisions in challenging flying conditions. The Boeing 747 is a wide-body aircraft that is widely utilized and has gained popularity for long-haul flights worldwide because of its exceptional capacity and extended-range capabilities. For many years, it has served as the backbone of large airlines, transporting millions of passengers and serving as a model of dependability and effectiveness for the aviation sector. In the cockpit, pilots primarily interact with the primary flight display. It provides navigational data and the status of the automated flying system in addition to digitally displaying fundamental flight characteristics like heading, altitude, and airspeed [2]. Because it is closely tied to the pilot's precise comprehension of key flight information, which has a direct and considerable impact on flight safety and efficiency, a thorough analysis of the Boeing 747 PFD interface is essential.

According to fluency theory research, it is difficult to discover a piece of information that is small or far away; if a piece of information is easily retrieved from memory, it creates an implicit awareness that it is easily retrievable, and this perception can arise during any cognitive process and can be used to inform subsequent judgments [3]. Fluency is essential in interface design since it not only minimizes user fatigue and error rates, but it also gives a better emotional experience on a psychological level. Fluency refers to the ability of users to accomplish activities smoothly and effortlessly, avoiding redundant steps and cognitive strain. Relevant research has demonstrated that interface design with high fluency can improve users' work efficiency, reduce users' cognitive strain, and provide users with a positive experience and trust [4]. Furthermore, fluency can encourage users to participate more actively in the operation, increasing their faith in the system. As a result, thorough research and the use of fluency design principles can not only

improve the user experience but also the usability of the interface and the satisfaction of users.

Meanwhile, as Artificial Intelligence (AI) continues to advance, an increasing number of researchers are bringing algorithmic models to the study of interface design. This includes the use of machine learning techniques to build models to acquire insight into user behavior, personalized needs, and interface interaction [5]. Researchers employ AI models to uncover critical characteristics such as user preferences, attention distribution, and cognitive load by analyzing vast amounts of user data in order to optimize interface design and improve the user experience. This trend not only delivers more accurate interface analysis tools, but it also encourages creativity and personalization of user interface design. With the incorporation of AI technology, interface design research is gradually moving in the direction of being smarter and closer to the demands of users.

In summary, this paper aims to investigate in detail the effects of PFD with varying fluency levels, gather experimental data to create a classification model, and offer researchers an intuitive method for evaluating and predicting interfaces through fluency theory-based interface discrimination.

2 MATERIALS AND METHODS

2.1 Fisher's Method of Discriminant Analysis

Based on the preceding explanation, the late modeling notion in this study is Fisher Discriminant Analysis. Fisher Discriminant Analysis (FDA) is a traditional statistical learning method that is mostly used for feature selection and data dimension reduction. To uncover traits that are better distinct between different classes, the strategy maximizes the between-class variation across different classes while minimizing the within-class variance within the same class [6]. The essential principle of Fisher's discrimination is to project the data into a new space, making samples from different categories more dispersed and samples within the same category more compact.This has resulted in a wide range of Fisher Discriminant applications in domains such as pattern recognition, face recognition, and bioinformatics.

In brief, the approach determines the best projection direction by maximizing inter-class variation and minimizing intra-class variance. The following are the key formulas for Fisher's discrimination, as well as the significance of each letter:

$$\mathrm{J}(\omega) = \frac{(\left(\omega^{\mathrm{T}}(\mu_1 - \mu_2)\right)^2}{\omega^{\mathrm{T}} \mathrm{S}_{\omega} \omega} \tag{1}$$

$\mathrm{J}(\omega)$ is the objective function of Fisher's discrimination, which aims to maximize the between-class variance and minimize the within-class variance; ω is the unit vector in the projection direction; μ_1 and μ_2 denote the mean vectors of the two classes respectively; and S_{ω} is the overall scattering matrix, which denotes the intraclass scattering matrix, including the dispersion of the data within the two classes.

2.2 Fluency Theory for Interfaces

Processing The ease or difficulty of information flow in the human cognitive system is referred to as fluency. Fluency theory focuses on improving the smoothness of the user experience in interface design and is specifically reflected in the optimization of interface features such as font size, color contrast, and graphic outlines [7]. According to relevant studies, designers can successfully guide users' attention and enhance the attractiveness of images by optimizing behaviors such as typefaces, colors, and typography [8]. These specific design ideas offer practical advice for boosting fluency and contribute to the creation of interfaces that better match user expectations.

2.3 Experimental Design

According to prior studies on fluency, the influencing aspects of fluency mostly include font size, color contrast, and graphic outline, among others. This experiment then creates three optimized interfaces based on the original Boeing 747 interface, each of which optimizes one influencing factor, two influencing factors, and three influencing factors. The volunteers observed the various interfaces and collected data for modeling and validation.

2.3.1 Experimental Purpose. To achieve the goal of scientific evaluation and prediction of the optimized redesigned interface, the Fisher discriminant model was constructed by combining the benefits of performance data, electrooculographic data, and subjective evaluation data. Specific data when subjects observed the interface was recorded.

2.3.2 Participants. The subjects were 16 Southeast University students (22-26 years old, 8 men and 8 women) who received flight simulator training, were healthy, had no sleep issues, and had normal or corrected vision. 1 hour before the trial, all volunteers were free of severe exercise and subjective discomfort.

2.3.3 Experimental Materials and Environment. The experimental scene was built using E-Prime software, and the experimental environment was kept calm during the experiment. After their heartbeats had stabilized, the individuals entered the laboratory, remained quietly for 5–10 minutes, and then began the experiment. As illustrated in Figure 1, the entire experiment exhibited four PFD pages (control interface, low fluency page, medium fluency page, and high fluency page). As shown in figure 2, the left half of the website displays the individual index parameters during the experiment, and the right side presents the optimization interface. Subjects must decide whether the left and right data are consistent and provide feedback. It should be emphasized that while judging each interface, there will be a blank page with the visual center point.

2.3.4 Experimental Procedure. The whole experiment consists of the following steps, as shown in Figure 3

1. The subjects began with a guided task consisting of going through the interface guides and introducing the experiment-related tasks; this phase was time-limited.

2. The subjects began to practice the experiment, which consisted of six random experimental interfaces (5000 ms), each followed by a blank page (5000 ms) with a center focal point.

3. After completing the activity and communicating with the primary subject, the participant clicked on the space to begin the official experiment. Each formal experiment needed subjects to determine if the data on 35 pages was consistent (700s).

Figure 1: From left to right, the original Boeing 747 PFD interface, the low-fluency optimized interface, the mid-fluency optimized interface, and the high-fluency optimized interface.

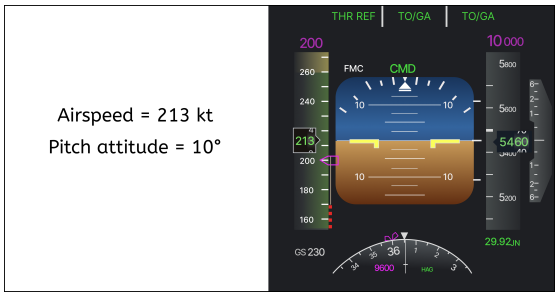

Figure 2: Experimental task interface: to determine whether the data on the left and right sides are the same.

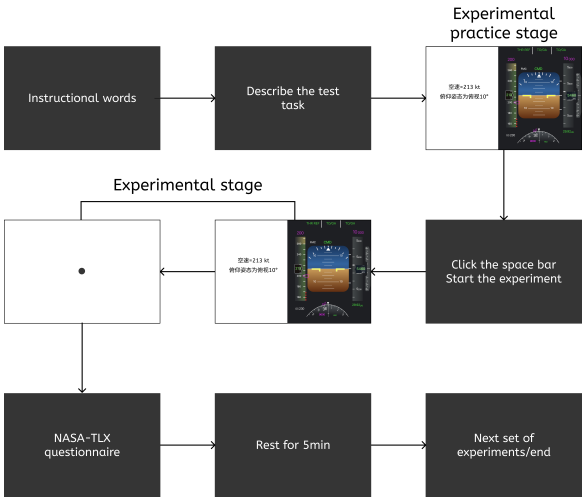

Figure 3: Experimental flow demonstration

4. Complete the matching NASA-TLX scale after the experiment, take a 5-minute rest, and then begin the following series of tests, making a total of four sets of tests (including the control group).

3 result

3.1 Analysis of Experimental Data

In the subjective evaluation based on the NASA-TLX scale, the results of a one-way repeated measures ANOVA showed that NASA-TLX presented horizontal significance ($F = 25.426$, $p<0.01$).

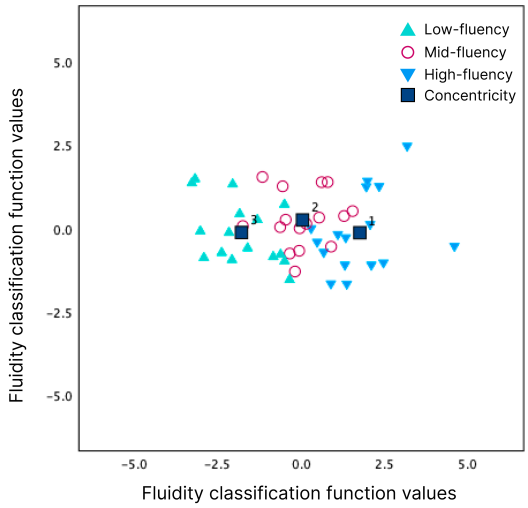

Figure 4: the multidimensional integrated discriminant prediction model

Among the electroocular evaluation indicators, the results of one-way repeated measures ANOVA showed that EOG presented horizontal significance ($F = 4.630$, $p<0.05$).

3.2 Fisher Model Building

Based on the results of the experimental measurements, a Fisher linear discriminant analysis method was used to construct a Boeing 747 cockpit PFD interface fluency discriminant model, which was used to optimize the determination of the interface fluency level. To ensure the comprehensiveness of the discrimination, the general discriminant analysis method is used. The discriminant model includes NASA-TLX subjective evaluation, reaction time, completion rate, and EOG.

Figure 4 displays the multidimensional integrated discriminant prediction model that was built through experimentation. Here is the discriminant model equation:

$$y_1 = 112.283x_1 + 20.12x_2 + 0.385x_3 + 1.012x_4 - 138.23 \quad (2)$$

$$y_2 = 119.068x_1 + 19.505x_2 + 0.49x_3 + 0.844x_4 - 132.233 \quad (3)$$

$$y_3 = 130.963x_1 + 18.094x_2 + 0.545x_3 + 0.686x_4 - 129.33 \quad (4)$$

Table 1: NASA-TLX ANOVA

"Category" (Mean ± Standard Deviation)				
"Medium fluency" (n=16)	"Low fluency" (n=16)	"High fluency" (n=16)	F	P
74.89±6.94	85.97±8.08	63.67±11.01	25.426	0.000**

Table 2: EOG ANOVA

"Category" (Mean ± Standard Deviation)				
"Medium fluency" (n=16)	"Low fluency" (n=16)	"High fluency" (n=16)	F	P
44.38±10.50	36.63±9.80	47.25±10.34	4.63	0.015*

Table 3: Wilks Lambda validity test

Wilk Lambda				
Function Check	Wilk Lambda	Chi-Square	Freedom	P
1-2	0.299	52.587	8	<0.01
2	0.967	1.455	3	<0.01

Table 4: Original data substitution test

Forecasting team member information					
Classification		1	2	3	Total
Count	1	13	13	0	16
	2	3	12	1	16
	3	0	4	12	16
%	1	81.3	18.8	0	100
	2	18.8	75	6.3	100
	3	0	25	75	100

Where y_1, y_2, and y_3 are the values of the discriminant function for the low-fluency, mid-fluency, and high-fluency interfaces, respectively. x_1 is the task correctness rate, x_2 is the reaction time for completing the task, x_3 is the value of the EOG electrooculographic metrics, and x_4 is the value of the NASA-TLX subjective evaluation score. Calculations were performed to compare the y_1, y_2, and y_3 values based on the values of x_1, x_2, x_3, and x_4, respectively. If the value of y_1 is the largest, the interface is considered to belong to the low-fluency classification; if the value of y_2 is the largest, the interface is classified as mid-fluency; if the value of y_3 is the largest, the interface is considered to be in the high-fluency classification.

The results of the Fisher discriminant function validity test are shown in the table. The original hypothesis of this test is that there is no significant difference in the mean Fisher Discriminant Function values of the different groups. From the p-value given in the table, $p<0.05$ indicates that the original hypothesis is justified to be rejected at the 0.05 level of significance.

The table gives the discriminant results, where 37 (out of 48) predictions by the discriminant function were correct, resulting in 77.1% of the original observations being judged correct. Considering the effectiveness of the above model validation, it is surmised that the model accuracy will be further improved by increasing the sample size at a later stage.

4 CONCLUSIONS

Interface recognition technology will become more mature as artificial intelligence continues to advance. According to the present development trend, this article focuses on the fluency influence elements on the interface of the Boeing 747, and investigates the impact of different fluencies on the subjects' browsing interface information. The interface's letter size, color contrast, and graphic contour are optimized, and a Fischer discriminant model is built for the design of interfaces with varying fluency. This method replaces the subjective emotional assessment that is most usually used in interface design evaluation by providing designers with a more intuitive review in a quantitative categorization style, providing a new idea of interface evaluation for future researchers.

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